

43.4 Smoothing splines

A P-spline starts from a basis and adds a penalty to it. The smoothing spline does the opposite: it asks a question about functions, with no basis in sight — among *all* twice-differentiable functions, which balances fit against roughness best? — and the answer turns out to be a spline, by the shortest and most satisfying derivation in the subject.

43.4.1 The variational problem

Measure the roughness of a function by $\int_a^b \{f''(t)\}^2 dt$: zero exactly for straight lines, large for a function that turns sharply, and, unlike the number of knots, defined for every function smooth enough to have it. Let

$$\mathcal{W} = \left\{ f : [a, b] \rightarrow \mathbb{R} \mid f' \text{ is absolutely continuous and } \int_a^b \{f''(t)\}^2 dt < \infty \right\}$$

be the set of such functions, and consider

$$S_\lambda(f) = \sum_{i=1}^n \{y_i - f(x_i)\}^2 + \lambda \int_a^b \{f''(t)\}^2 dt, \quad \lambda > 0. \quad (43.4.1)$$

Take the design points distinct and ordered, $a \leq x_1 < \cdots < x_n \leq b$. (If some coincide, replace each group by its mean with a weight equal to its size; [Exercise \(B1\)](#) checks that this changes nothing.)

The two extremes are clear. As $\lambda \rightarrow 0$ the criterion is minimized by any function through the data, so the problem is ill-posed without the penalty; as $\lambda \rightarrow \infty$ the penalty forces $f'' \equiv 0$ and the minimizer is the least squares line. In between the minimizer is some compromise between an interpolating curve and a line, and it is not obvious in advance that it is finite-dimensional.

Definition 43.4.1. Natural cubic spline

A **natural cubic spline** with knots $x_1 < \cdots < x_n$ is a function $g : [a, b] \rightarrow \mathbb{R}$ that is a cubic polynomial on each $[x_i, x_{i+1}]$, has two continuous derivatives on $[a, b]$, and is linear on $[a, x_1]$ and on $[x_n, b]$. Equivalently: on $[x_1, x_n]$, g is a cubic spline with interior knots $x_2, \dots, x_{n-1}$ satisfying the **natural boundary conditions** $g''(x_1) = g''(x_n) = 0$, extended to $[a, b]$ by its tangent lines at x_1 and x_n . (The two conditions are exactly what makes the extension C^2 .)

43.4.2 Interpolation with the least roughness

Everything follows from one integration by parts.

Lemma 43.4.1. The natural interpolant, and why it is the smoothest

Let $n \geq 2$ and $x_1 < \cdots < x_n$.

- (a) If g is a natural cubic spline with these knots and $h \in \mathcal{W}$ satisfies $h(x_i) = 0$ for every i , then $\int_a^b g'' h'' = 0$.
- (b) For every $\mathbf{z} \in \mathbb{R}^n$ there is exactly one natural cubic spline g with $g(x_i) = z_i$; the space of natural cubic splines with these knots has dimension n .

- (c) That g is the unique minimizer of $\int_a^b f''^2$ among all $f \in \mathcal{W}$ with $f(x_i) = z_i$.

Proof.

- (a) Integrating by parts, $\int_a^b g'' h'' = [g'' h']_a^b - \int_a^b g''' h'$. The boundary term vanishes because g is linear near a and b , so $g''(a) = g''(b) = 0$. On $[a, x_1]$ and $[x_n, b]$ we have $g''' = 0$, and on (x_i, x_{i+1}) the third derivative of a cubic is a constant c_i . Hence

$$\int_a^b g''' h' = \sum_{i=1}^{n-1} c_i \int_{x_i}^{x_{i+1}} h' = \sum_{i=1}^{n-1} c_i \{h(x_{i+1}) - h(x_i)\} = 0.$$

- (b) A cubic spline on $[x_1, x_n]$ with the $n - 2$ interior knots $x_2, \dots, x_{n-1}$ lives in a space of dimension $4 + (n - 2) = n + 2$ (Theorem 42.3.1 with $d = 3$), and extending it linearly outside is determined once $g''(x_1) = g''(x_n) = 0$. Those are two linear constraints, so the natural cubic splines form a space of dimension at least n . The evaluation map $g \mapsto \{g(x_1), \dots, g(x_n)\}$ is linear into $\mathbb{R}^n$; if g is in its kernel, apply (a) with $h = g$ to get $\int_a^b g''^2 = 0$, so $g'' \equiv 0$ and g is linear, and a line vanishing at $n \geq 2$ distinct points is zero. The map is therefore injective, so the dimension is at most n and hence exactly n , and the map is a bijection.
- (c) Let $f \in \mathcal{W}$ interpolate $\mathbf{z}$ and put $h = f - g$, so $h(x_i) = 0$. By (a) the cross term vanishes, leaving

$$\int_a^b f''^2 = \int_a^b g''^2 + 2 \int_a^b g'' h'' + \int_a^b h''^2 = \int_a^b g''^2 + \int_a^b h''^2,$$

which is at least $\int_a^b g''^2$, with equality iff $h'' \equiv 0$, that is iff h is linear; and a line vanishing at $n \geq 2$ points is zero. ■

Theorem 43.4.2. The smoothing spline

Let $\lambda > 0$ and $n \geq 2$ distinct design points. Then S_λ of Eq. 43.4.1 has a unique minimizer over $\mathcal{W}$, and it is the natural cubic spline with knots at $x_1, \dots, x_n$ whose vector of values is

$$\hat{\mathbf{f}} = (\mathbf{I} + \lambda \mathbf{P})^{-1} \mathbf{y}, \quad (43.4.2)$$

where $\mathbf{P}$ is the nonnegative definite matrix representing the roughness, $\int_a^b g''^2 = \mathbf{z}^\top \mathbf{P} \mathbf{z}$, of the natural cubic spline with values $\mathbf{z}$.

Proof. Let $f \in \mathcal{W}$ be arbitrary and let g be the natural cubic spline with $g(x_i) = f(x_i)$, which exists and is unique by Lemma 43.4.1(b). The two functions have the same residual sum of squares, and by Lemma 43.4.1(c) the penalty of g is no larger, strictly smaller unless $f = g$. Hence $S_\lambda(g) \leq S_\lambda(f)$ with equality only if $f = g$, and it suffices to minimize over the n -dimensional space of natural cubic splines.

Parameterize that space by $\mathbf{z} \in \mathbb{R}^n$ through the bijection of Lemma 43.4.1(b). Since g depends linearly on $\mathbf{z}$, so does g'' , and the penalty $\int_a^b g''^2$ is a quadratic form $\mathbf{z}^\top \mathbf{P} \mathbf{z}$ with $\mathbf{P}$ symmetric and nonnegative definite. Eq. 43.4.1 becomes $\|\mathbf{y} - \mathbf{z}\|^2 + \lambda \mathbf{z}^\top \mathbf{P} \mathbf{z}$, a strictly convex

quadratic in $\mathbf{z}$ because $\mathbf{I} + \lambda \mathbf{P}$ is positive definite; by Proposition 1.11.2 its unique minimizer is Eq. 43.4.2. ■

Idea. The theorem is a *representer* result: an optimization over an infinite-dimensional space of functions has a finite-dimensional solution, with one basis function per observation. The reason is the orthogonality in Lemma 43.4.1(a): any direction in which f differs from a natural spline without changing the fit costs roughness and buys nothing. Kimeldorf and Wahba (1971) proved the general version; see the notes.

43.4.3 The Reinsch algorithm

The proof used $\mathbf{P}$ without exhibiting it. Doing so turns Eq. 43.4.2, an $n \times n$ dense solve, into a banded one.

Proposition 43.4.3. Band matrices for the roughness

Let $h_i = x_{i+1} - x_i$ for $i = 1, \dots, n-1$, and let $n \geq 3$. Define $\mathbf{Q}$, of size $n \times (n-2)$, and $\mathbf{R}$, symmetric tridiagonal of order $n-2$, by

$$q_{j-1,j} = \frac{1}{h_{j-1}}, \quad q_{j,j} = -\frac{1}{h_{j-1}} - \frac{1}{h_j}, \quad q_{j+1,j} = \frac{1}{h_j},$$

$$r_{jj} = \frac{h_{j-1} + h_j}{3}, \quad r_{j,j+1} = r_{j+1,j} = \frac{h_j}{6},$$

for $j = 2, \dots, n-1$ (columns and rows indexed by the interior knots), all other entries zero. Then:

- (a) $\mathbf{R}$ is positive definite.
- (b) If g is the natural cubic spline with values $\mathbf{z}$ and $\boldsymbol{\gamma} = \{g''(x_2), \dots, g''(x_{n-1})\}^\top$, then $\mathbf{R}\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{z}$ and $\int_a^b g''^2 = \boldsymbol{\gamma}^\top \mathbf{R} \boldsymbol{\gamma}$, so that $\mathbf{P} = \mathbf{Q} \mathbf{R}^{-1} \mathbf{Q}^\top$.
- (c) The fit of Theorem 43.4.2 satisfies

$$(\mathbf{R} + \lambda \mathbf{Q}^\top \mathbf{Q}) \hat{\boldsymbol{\gamma}} = \mathbf{Q}^\top \mathbf{y}, \quad \hat{\mathbf{f}} = \mathbf{y} - \lambda \mathbf{Q} \hat{\boldsymbol{\gamma}}, \quad (43.4.3)$$

and $\mathbf{R} + \lambda \mathbf{Q}^\top \mathbf{Q}$ has bandwidth two, so the fit costs $O(n)$ operations.

Proof.

- (a) $\mathbf{R}$ is symmetric with positive diagonal, and $r_{j,j-1} + r_{j,j+1} = (h_{j-1} + h_j)/6 < r_{jj}$, so it is strictly diagonally dominant and therefore positive definite.
- (b) On $[x_j, x_{j+1}]$, g'' is linear with endpoint values γ_j and γ_{j+1} (setting $\gamma_1 = \gamma_n = 0$). Writing $u = t - x_j$ and integrating twice,

$$g(x_j + u) = z_j + g'(x_j)u + \frac{1}{2}\gamma_j u^2 + \frac{\gamma_{j+1} - \gamma_j}{6h_j} u^3.$$

Setting $u = h_j$ and solving for $g'(x_j)$ gives $g'(x_j) = (z_{j+1} - z_j)/h_j - h_j(2\gamma_j + \gamma_{j+1})/6$,

and differentiating at $u = h_j$,

$$g'(x_{j+1}^-) = \frac{z_{j+1} - z_j}{h_j} + \frac{h_j(\gamma_j + 2\gamma_{j+1})}{6}.$$

Equating this with $g'(x_{j+1}^+)$, read off the interval $[x_{j+1}, x_{j+2}]$, and writing $i = j + 1$,

$$\frac{z_{i+1} - z_i}{h_i} - \frac{z_i - z_{i-1}}{h_{i-1}} = \frac{h_{i-1}\gamma_{i-1} + 2(h_{i-1} + h_i)\gamma_i + h_i\gamma_{i+1}}{6},$$

which is $(\mathbf{Q}^\top \mathbf{z})_i = (\mathbf{R}\boldsymbol{\gamma})_i$. For the penalty, $\int_{x_j}^{x_{j+1}} g''^2 = h_j(\gamma_j^2 + \gamma_j\gamma_{j+1} + \gamma_{j+1}^2)/3$ by direct integration of a linear function squared, and summing over j reproduces $\boldsymbol{\gamma}^\top \mathbf{R}\boldsymbol{\gamma}$ term by term. Since $\boldsymbol{\gamma} = \mathbf{R}^{-1}\mathbf{Q}^\top \mathbf{z}$, the penalty is $\mathbf{z}^\top \mathbf{Q}\mathbf{R}^{-1}\mathbf{Q}^\top \mathbf{z}$.

- (c) Put $\boldsymbol{\gamma} = \mathbf{R}^{-1}\mathbf{Q}^\top \hat{\mathbf{f}}$. From [Eq. 43.4.2](#), $\hat{\mathbf{f}} + \lambda \mathbf{Q}\boldsymbol{\gamma} = \mathbf{y}$, and multiplying by $\mathbf{Q}^\top$ and using $\mathbf{R}\boldsymbol{\gamma} = \mathbf{Q}^\top \hat{\mathbf{f}}$ gives $\mathbf{R}\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{y} - \lambda \mathbf{Q}^\top \mathbf{Q}\boldsymbol{\gamma}$, which is [Eq. 43.4.3](#). $\mathbf{R}$ is tridiagonal and $\mathbf{Q}$ has three nonzero entries per column in consecutive rows, so $\mathbf{R} + \lambda \mathbf{Q}^\top \mathbf{Q}$ has bandwidth two. A Cholesky factor inherits the bandwidth: in [Eq. 10.2.1](#), if $r_{ki} = 0$ whenever $i - k > 2$ for every $k < j$, then for $i - j > 2$ every product $r_{kj}r_{ki}$ in the sum vanishes (a nonzero r_{kj} needs $j - k \leq 2$, whence $i - k > 2$) and so does a_{ji} , so $r_{ji} = 0$. The factorization of [Theorem 10.2.1](#) therefore visits $O(n)$ entries instead of $O(n^3)$, and so do the two multiplications by $\mathbf{Q}$. ■

Reinsch (1967) gave this algorithm before the variational theorem was widely known; it is why smoothing splines remain the cheapest smoother for a single covariate. Nothing in it forms $\mathbf{S}_\lambda = (\mathbf{I} + \lambda \mathbf{P})^{-1}$, which is dense; the diagonal of $\mathbf{S}_\lambda$, needed for cross-validation in [Section 43.5](#), can also be got in $O(n)$ from a banded inverse (Hutchinson and de Hoog, 1985).

Example. A smoothing spline, checked three ways

Take $n = 120$ design points drawn uniformly on $[0, 1]$ and the mean function of [Example](#) (Bias, variance and the optimal bandwidth) with $\sigma = 0.25$. At $\lambda = 1 \times 10^{-4}$ the fit uses 12.32 effective degrees of freedom. Three independent computations agree with it. The roughness $\boldsymbol{\gamma}^\top \mathbf{R}\boldsymbol{\gamma}$ of the natural spline interpolating the true curve, 1069.5, matches $\int_a^b g''^2$ from a separately constructed natural cubic interpolant to a relative error of 1.6×10^{-12} ; the banded solve [Eq. 43.4.3](#) agrees with the dense form [Eq. 43.4.2](#) to 1.3×10^{-10} ; and minimizing $\|\mathbf{y} - \mathbf{B}\boldsymbol{\gamma}\|^2 + \lambda \boldsymbol{\gamma}^\top \boldsymbol{\Omega}\boldsymbol{\gamma}$ over the *whole* space of cubic splines with a knot at every design point — dimension 122, with $\boldsymbol{\Omega}_{jk} = \int B_j'' B_k''$ computed exactly — gives fitted values differing from the smoothing spline by 4.4×10^{-10} . That is [Theorem 43.4.2](#) in numerical form: the minimizer over the larger space is the natural spline.

```
import numpy as np

def f_true(x):
    """The mean function of Section 43.1."""
    return 2 * x + np.exp(-25 * (x - 0.35) ** 2) - 0.75 * np.exp(-50 * (x - 0.80) ** 2)

def reinsch(t):
```

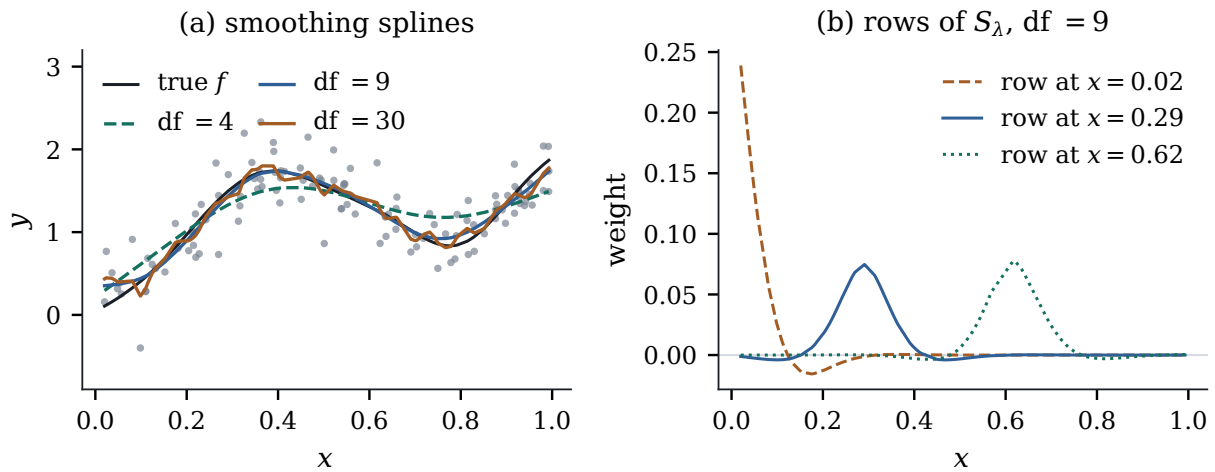


Figure 43.4.1: (a) Smoothing splines at three values of λ , labelled by effective degrees of freedom. (b) Three rows of S_λ at $df = 9$, the weights producing $\hat{f}(x)$ at $x = 0.02, 0.29$ and 0.62 : local, summing to one, slightly negative beside the peak.

```

"""The band matrices Q (N x (N-2)) and R ((N-2) x (N-2)) of the Reinsch form."""
N = len(t)
h = np.diff(t)
Q = np.zeros((N, N - 2))
R = np.zeros((N - 2, N - 2))
for j in range(N - 2):
    # column j belongs to interior knot
    t[j+1]
    Q[j, j], Q[j + 1, j], Q[j + 2, j] = 1 / h[j], -1 / h[j] - 1 / h[j + 1], 1 /
    h[j + 1]
    R[j, j] = (h[j] + h[j + 1]) / 3
    if j + 1 < N - 2:
        R[j, j + 1] = R[j + 1, j] = h[j + 1] / 6
return Q, R

def smoothing_spline(t, y, lam):
    """Fitted values of the smoothing spline, by solving a pentadiagonal system."""
    Q, R = reinsch(t)
    gamma = np.linalg.solve(R + lam * Q.T @ Q, Q.T @ y) # the second derivatives
    return y - lam * Q @ gamma, gamma

def smoother_matrix(t, lam):
    """The matrix S_lambda with fitted values S_lambda y (formed only to look at it)"""
    Q, R = reinsch(t)
    return np.eye(len(t)) - lam * Q @ np.linalg.solve(R + lam * Q.T @ Q, Q.T)

n, sigma = 120, 0.25
rng = np.random.default_rng(430430)
t = np.sort(rng.uniform(0, 1, n))
y = f_true(t) + sigma * rng.normal(size=n)

lam = 1e-4
fit, _ = smoothing_spline(t, y, lam)
S = smoother_matrix(t, lam)

```

```
df = np.trace(S)
print("effective degrees of freedom at lambda = 1e-4:", round(df, 2))
print("rows of S sum to one, largest departure:", np.max(np.abs(S.sum(axis=1) - 1)))
```

[Run this code online](#)

43.4.4 Variations, and the relation to penalized splines

Other orders. Replacing f'' by $f^{(m)}$ in Eq. 43.4.1 gives a **natural spline of degree $2m-1$** : the proof of Lemma 43.4.1 goes through with m integrations by parts and the boundary conditions $f^{(j)}(x_1) = f^{(j)}(x_n) = 0$ for $j = m, \dots, 2m-2$. The case $m = 1$ gives a piecewise linear fit, $m = 2$ the cubic one above, $m = 3$ a quintic. In practice $m = 2$ is used almost always; the extra smoothness of $m = 3$ rarely shows.

Reproducing kernels. The penalty $\int f''^2$ is the squared norm of the part of f orthogonal to the straight lines, in a Hilbert space of functions in which evaluation $f \mapsto f(x)$ is a bounded linear functional and so representable as an inner product with a function R_x , the *reproducing kernel*. In that language Theorem 43.4.2 is a special case of the representer theorem of Kimeldorf and Wahba (1971): the minimizer of $\sum_i \{y_i - f(x_i)\}^2 + \lambda \|Pf\|^2$ lies in the span of $R_{x_1}, \dots, R_{x_n}$ plus the null space of P . That is the route of Wahba (1990), and it covers thin-plate splines, splines on a sphere and the kernel methods of machine learning. This book does not develop it; the proof above is self-contained for a single covariate.

Smoothing splines and P-splines. A smoothing spline is a penalized spline with a knot at every distinct design point and the exact derivative penalty, as Example (A smoothing spline, checked three ways) confirms numerically, and a P-spline with $K = 20$ knots is a *low-rank* approximation to it: both fits are determined, through Eq. 43.3.2, by how the penalty ranks a handful of smooth directions, so the two agree once m comfortably exceeds the effective degrees of freedom. The smoothing spline has no knot choice and costs $O(n)$; the P-spline carries an explicit coefficient vector and generalizes to several covariates and to nonnormal responses, which is why Chapter 44 uses it.

Remark (The equivalent kernel of a smoothing spline). Panel (b) of Figure 43.4.1 shows the rows of $\mathbf{S}_\lambda$, the smoothing spline's equivalent kernel in the sense of Section 43.1. Silverman (1984) proved that at an interior point, for large n and small λ , the weight given to a design point at distance u is approximately $\{nb(x)g(x)\}^{-1} \frac{1}{2} \exp(-|v|/\sqrt{2}) \sin(|v|/\sqrt{2} + \pi/4)$ with $v = u/b(x)$ and local bandwidth $b(x) = \{\lambda/(ng(x))\}^{1/4}$. Two features of that formula show in the picture: the weights decay exponentially, and the sine makes them change sign, so a smoothing spline is a kernel smoother with a small negative side lobe — here the most negative weight is -0.0034 against a peak of 0.0778 . The bandwidth adapts to the design density by itself, as $g(x)^{-1/4}$. We state this without proof; the argument is a Green's-function calculation.

43.4.5 Exercises

43.4.5.1 A. Check your understanding

Exercise. A1

Show directly from Eq. 43.4.3 that $\hat{\mathbf{f}} \rightarrow \mathbf{y}$ as $\lambda \rightarrow 0$ and that $\hat{\mathbf{f}}$ tends to the least squares line

as $\lambda \rightarrow \infty$. (For the second, use [Proposition 43.4.3\(b\)](#): what does $\mathbf{Q}^\top \mathbf{z} = \mathbf{0}$ say about $\mathbf{z}$?)

Solution. As $\lambda \rightarrow 0$, [Eq. 43.4.3](#) gives $\hat{\mathbf{y}} \rightarrow \mathbf{R}^{-1} \mathbf{Q}^\top \mathbf{y}$, bounded, so $\hat{\mathbf{f}} = \mathbf{y} - \lambda \mathbf{Q} \hat{\mathbf{y}} \rightarrow \mathbf{y}$. As $\lambda \rightarrow \infty$, the penalty $\mathbf{z}^\top \mathbf{P} \mathbf{z}$ must tend to zero, and $\mathbf{P} = \mathbf{Q} \mathbf{R}^{-1} \mathbf{Q}^\top$ with $\mathbf{R}$ positive definite, so $\mathbf{Q}^\top \mathbf{z} \rightarrow \mathbf{0}$. Now $(\mathbf{Q}^\top \mathbf{z})_i$ is a second divided difference of $\mathbf{z}$ at x_{i-1}, x_i, x_{i+1} , and it vanishes for all i exactly when the points (x_i, z_i) are collinear. So the limit lies in the two-dimensional space $\mathcal{C}([1, \mathbf{x}])$, and since the residual sum of squares is still being minimized there, it is the least squares line.

43.4.5.2 B. Practice

Exercise. B1

Suppose the design has ties: the distinct values are $t_1 < \dots < t_N$ and n_k observations share t_k , with mean $\bar{y}_k$. Show that $\sum_i \{y_i - f(x_i)\}^2 = \sum_k n_k \{\bar{y}_k - f(t_k)\}^2 + C$ with C free of f , so that [Theorem 43.4.2](#) holds with knots at the distinct values and $\mathbf{I}$ replaced by $\text{diag}(n_1, \dots, n_N)$ in [Eq. 43.4.2](#).

Exercise. B2

Using the cell above, build $\mathbf{Q}$ and $\mathbf{R}$ for $n = 8$ equally spaced points on $[0, 1]$ and verify by hand that $\mathbf{R}$ is tridiagonal with $r_{jj} = 2h/3$ and $r_{j,j+1} = h/6$, and that $\mathbf{Q}^\top \mathbf{1} = \mathbf{0}$ and $\mathbf{Q}^\top \mathbf{x} = \mathbf{0}$. What does the second pair of identities say about $\mathbf{S}_\lambda \mathbf{1}$ and $\mathbf{S}_\lambda \mathbf{x}$?

43.4.5.3 C. Going deeper

Exercise. C1

State and prove the analogue of [Lemma 43.4.1\(a\)](#) for the penalty $\int_a^b \{f^{(m)}\}^2$, with g a natural spline of degree $2m - 1$. Which boundary conditions are needed to kill the boundary terms after m integrations by parts, and what replaces “a line vanishing at $n \geq 2$ points is zero”?

Exercise. C2

Show that if $\lambda > 0$ then $\hat{\mathbf{f}}$ never interpolates the data unless the data already lie on a straight line. (Use [Eq. 43.4.2](#) and the fact that $\mathbf{P} \mathbf{z} = \mathbf{0}$ iff $\mathbf{z} \in \mathcal{C}([1, \mathbf{x}])$, which [Exercise \(A1\)](#) establishes.) Contrast this with a regression spline with $K = n - 4$ knots, and say what the difference means for the leave-one-out shortcut of [Theorem 29.3.2](#).

Solution. $\hat{\mathbf{f}} = \mathbf{y}$ requires $(\mathbf{I} + \lambda \mathbf{P}) \mathbf{y} = \mathbf{y}$, that is $\mathbf{P} \mathbf{y} = \mathbf{0}$, that is $\mathbf{y}$ in $\mathcal{C}([1, \mathbf{x}])$. So for $\lambda > 0$ the fit leaves a nonzero residual unless the data are exactly linear, and every s_{ii} is strictly less than one, which is what makes the shortcut of [Theorem 29.3.2](#) well defined. An unpenalized regression spline with n basis functions interpolates, every $s_{ii} = 1$, the shortcut divides by zero, and cross-validation is useless — one more reason the dial should be a penalty and not a knot count.